

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Pharmacy SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Pharmacy Branch : Pharmacy

Semester : SEMESTER - 8

Subject Code & Name: ET - 1 - BP810ET - EXPERIMENTAL PHARMACOLOGY

Time : 3 Hours]

[Total Marks : 75

Instructions to the Students:

- 1.All questions are compulsory.
- 2.Draw diagrams / figures wherever necessary
- 3.Figures to right indicate full marks

Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

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|---|---|
| A) Define CPCSEA. | 2 |
| B) What is meant by OECD guidelines? | 2 |
| C) Name any two common laboratory animals used in preclinical studies. | 2 |
| D) What is a transgenic animal? | 2 |
| E) Mention any two routes of drug administration in laboratory animals. | 2 |
| F) Define euthanasia. | 2 |
| G) What is a sham control group? | 2 |
| H) Define dose conversion between species. | 2 |
| I) What is Student's t-test used for? | 2 |
| J) Mention any two applications of graphical representation of data. | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

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|---|----|
| A) Discuss preclinical screening models for CNS activity including analgesic, antipyretic, sedative-hypnotic and antiepileptic drugs. | 10 |
| B) Describe preclinical screening models for CVS activity including antihypertensives, diuretics and antiarrhythmic drugs. | 10 |
| C) Describe preclinical data analysis using Student's t-test and One-way ANOVA with suitable examples and graphical representation. | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

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| A) Describe common laboratory animals with their applications. | 5 |
| B) Explain techniques for blood collection in laboratory animals. | 5 |
| C) Explain dose selection and preparation of drug solutions/suspensions. | 5 |
| D) Write a short note on screening models for antidiabetic drugs. | 5 |
| E) Explain screening models for analgesic and antipyretic activity. | 5 |
| F) Describe screening models for anti-Parkinson's drugs. | 5 |
| G) Write a short note on screening models for antiasthmatic drugs. | 5 |
| H) Explain CPCSEA guidelines for maintenance and breeding of laboratory animals. | 5 |
| I) Write a note on popular transgenic and mutant animals. | 5 |

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